1. Run report: EXP-ECDLP-8c41d7 — m=4 block order, L=2..4

Observations only. This report records what ran and what was measured. It has no verdict, support or weaken language; those belong to the evidence review.

- **Experiment** EXP-ECDLP-8c41d7 · **hypothesis** H-ECDLP-60de36 · **question** RQ-ECDLP-001 · **goal** GOAL-ECDLP-001 · **protocol** `EXP-ECDLP-8c41d7/specification.yaml` with its amendments.
- **Host** 4 vCPU Intel Xeon 2.10 GHz, 15 GB; msolve 0.6.5-1build2 (1 thread per process); Singular 4.3.2. Four worker processes, one solver process per solve. Timeout 600 s per solve. Seeds: sha256 of `EXP-ECDLP-f49f75/v1|m|L|instance`.

1.1. Runs

run	started (UTC)	finished	rows	git HEAD
8c41d7-01	2026-10-16 01:07	none (lost)	6	6d6df97b6d
8c41d7-02	2026-10-16 03:15	none (lost)	6	04a87943e2
8c41d7-03	2026-10-19 16:58	2026-10-19 17:38	6	4ad12014e3
8c41d7-04	2026-10-19 17:41	2026-10-19 18:32	6	7413b95b6a

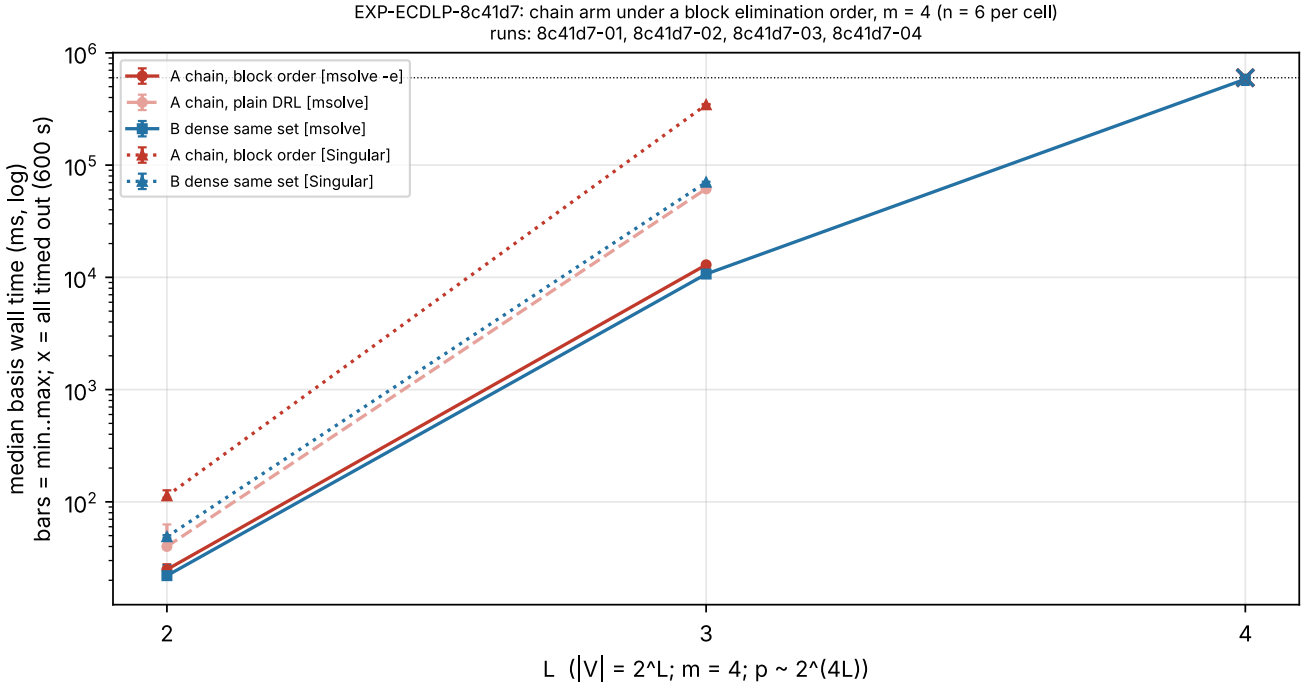


Figure 1: Median basis time per arm vs L (measured; process wall time), bars min–max over completed instances; × = every instance timed out (600 s). Source runs as in the table. Where a cell mixes two primes (EXP-ECDLP-8c41d7, L = 4: Singular arms at p = 65551, msolve arms at p = 262271) only comparisons within one solver family are on shared instances.

1.2. Per-cell measurements

Columns: completed / timed out / other failure / degenerate target; median ms [min–max]; planted-in-variety (passed/checked); max F4 degree (msolve arms). A cell can mix two runs (and two primes, `p bits`) where an amendment re-ran arms; the source runs are listed.

L	p bits	arm	runs	ok/to/oth/deg	median ms [min–max]	planted	F4 deg
2	9	A_block_sg	01	6/0/0/0	113 [109–127]	4/4	—
2	9	A_drl_ms	01	6/0/0/0	40 [38–63]	4/4	32
2	9	A_elim_ms	01	6/0/0/0	25 [23–28]	4/4	32
2	9	B_ms	01	6/0/0/0	22 [20–27]	4/4	32
2	9	B_sg	01	6/0/0/0	49 [48–50]	4/4	—
3	13	A_block_sg	02	5/0/0/1	344,621 [344,433–349,883]	3/3	—
3	13	A_drl_ms	02	5/0/0/1	61,421 [59,168–62,017]	3/3	32
3	13	A_elim_ms	02	5/0/0/1	12,929 [11,990–13,389]	3/3	32

3	13	B_ms	02	5/0/0/1	10,700 [10,378–10,976]	3/3	32
3	13	B_sg	02	5/0/0/1	70,364 [65,931–71,061]	3/3	—
4	17	A_block_sg	03	0/5/0/1	— [——]	—	—
4	19	A_drl_ms	04	0/4/1/1	— [——]	—	—
4	19	A_elim_ms	04	0/5/0/1	— [——]	—	—
4	19	B_ms	04	5/0/0/1	582,996 [518,230–586,271]	3/3	40
4	17	B_sg	03	0/5/0/1	— [——]	—	—

1.3. Operational notes (facts, not findings)

- Three interruptions: RUN-01 (L = 2 complete) lost to a VM restart during L = 3; RUN-02 (L = 3 complete) lost to a second VM restart during L = 4; RUN-03 (L = 4) ran to the end. Amendments AMD-01 and AMD-02 resumed the unexecuted cells with an unchanged launcher, seeds, arms, timeout and stopping rule.
- RUN-03, cell L = 4 at p = 65551 (17 bits): every msolve arm crashed with a segmentation fault (rc −11) on every instance within a fraction of a second; both Singular arms timed out on all six instances. The crash is an msolve 0.6.5 defect on 17- and 18-bit primes (reproduced on standalone systems; CORR-20261019-e6f65b, AMD-03).
- RUN-04 (AMD-03) re-ran the msolve arms of L = 4 at p = 262271 (19 bits), via block_order_m4_p19.py, which overrides only that prime choice. The Singular arms of L = 4 stay at p = 65551 from RUN-03. RUN-03′s msolve rows are instrument failures and are replaced by RUN-04′s in the merged table.
- RUN-04, cell L = 4: B_ms completed on every non-degenerate instance in 518–586 s (cap 600 s); A_elim_ms timed out on every one; A_drl_ms timed out on all but one instance, which was killed with rc −9 after 184 s (signal 9; cause not diagnosed. The validator measured A_drl_ms alone reaching 3.4 GB by 200 s, so four concurrent copies may have exceeded the memory budget; unproven). One of six instances had a degenerate planted target (sum = point at infinity).
- A unit-test run (about 6 s on one core, including a msolve canary) shared the host with RUN-04 at about 17:44 UTC (recorded in its operational-notes.txt).
- Every planted decomposition found lies in the computed basis; the eliminated chain basis equals the dense basis on every instance where both finished (L = 2, 3).